

DP Computer Science: A 1.1.8 Compression

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- 1. Define **compression** and state why it is used, in terms of reducing file size by removing **redundancy**
- 2. Distinguish **lossless** from **lossy** compression by their defining property: exact reconstruction against permanent discard of data
- 3. Describe **run-length encoding** and apply it to a short string to produce the encoded form, then decode it back
- 4. Describe **transform coding** as the conversion of data into the frequency domain and the removal of imperceptible detail, using **JPEG** or **MP3** as the example
- 5. Select a lossless or lossy method for a given file type and justify the choice

Command Term Note: The lesson is held at **Describe**—a detailed account. It stops there. Evaluation, algorithm design, and ethical dilemmas have been cut because they do not earn marks at this command term.

Key Vocabulary (Tier 3)

Term	Definition
Compression	Representing the same information in fewer bits so a file takes less space
Redundancy	Information that is repeated or predictable, and can be removed without losing meaning
Lossless compression	Compression from which the original file is rebuilt exactly
Lossy compression	Compression that permanently discards data judged imperceptible, so the original cannot be rebuilt

Term	Definition
Run-length encoding (RLE)	A lossless method that replaces a run of repeated values with a count and the value
Transform coding	A lossy method that converts data into frequencies and discards the components people cannot perceive

Pedagogical Focus

- **Thinking Skills (ATL):** Students consciously practise **classification and comparison**—classifying file types as suited to lossless or lossy compression and justifying each placement, and reasoning about why RLE shortens some data but lengthens data with no repeats.
- **Conceptual Understanding (ATT):** The lesson is built to move from **concrete to abstract** rather than opening with a definition. The order of teaching is **explain then apply**—teach the concept and the worked example first, and give independent practice only once the idea is secure.

Name the skill to students: *"Today you are practising classifying and justifying, not just recalling."*

Secondary ATLs present: Communication skills (through precise written description) and self-management skills (through following the encode and decode procedure accurately).

EAL Scaffold

Support Type	Description
Sentence Frame (Comparison)	"_____ compression is [lossless/lossy], because _____. This means the original file _____."
Sentence Frame (Difference)	"One difference between lossless and lossy compression is that _____, whereas _____."
Word Bank	For lossless/lossy comparison

2. Lesson Sequence (One-Hour Block)

Part 1: Do Now (Retrieval) — ~10%

Activity: Three short recall questions from earlier in the syllabus sequence:

Question	Source
1. State the number of bits in a byte, and name one unit larger than a byte	[A1.1.1]
2. Name one type of primary memory and state what it holds	[A1.1.4]
3. Give one reason a computer needs secondary or persistent storage	[A1.1.7]

Part 2: Main Sequence — ~60%

Phase 1: Explicit Teaching from a Concrete Example (~10 mins)

Step 1 — The Concrete Example:

| "A four-minute song drops from about 40 MB to about 4 MB. How?"

Define Compression:

| Representing the same information in fewer bits so a file takes less space.

Define Redundancy:

| Information that is repeated or predictable, and can be removed without losing meaning.

Step 2 — Lossless vs. Lossy (Second Concrete Pairing):

File Type	Compression Type	Why?
Python program (.py)	Lossless	One wrong byte stops it running; original must be rebuilt exactly
Holiday photo (.jpg)	Lossy	Small loss of detail goes unnoticed

Establish the Defining Property of Each:

Type	Defining Property
Lossless	Rebuilds the original exactly
Lossy	Discards data permanently; original cannot be rebuilt

Phase 2: Progressive Abstraction Through the Two Techniques (~15 mins)

Technique 1: Run-Length Encoding (RLE) — Lossless

Worked Example (on the board):

Original	Encoded
AAAABBBCCDAA (12 characters)	4A3B2C1D2A (10 characters)

Step 1 — Encode: Count consecutive repeated characters and write the count followed by the character.

Step 2 — Decode: Take each count-character pair and expand it back to the original.

| *"The original returns unchanged."*

The Failure Case (Showing the Limit):

Original	Encoded	Result
ABCD (4 characters)	1A1B1C1D (8 characters)	Doubles in size!

| *"RLE shortens repeated data but lengthens data with no repeats. Compression is not guaranteed to shrink; it depends on the data."*

Technique 2: Transform Coding — Lossy

Conceptual Explanation:

Step	Description
1. Convert	Data is converted into frequencies (frequency domain)
2. Discard	Components people cannot perceive are removed
3. Store	The remaining data is stored (smaller file)

Examples:

File Type	Transform Coding Example
Images	JPEG
Audio	MP3

Close the Phase — Name the General Principle:

| Both techniques remove redundancy. They differ on *whether the removed data can be recovered*.

	RLE	Transform Coding
Type	Lossless	Lossy
What it removes	Repeated characters/patterns	Imperceptible frequencies
Can original be rebuilt?	Yes	No

Phase 3: Linking Task (~5 mins)

Task: Students build a short comparison or concept map connecting compression back to earlier topics:

Connection	Why It Matters
Secondary Storage (A1.1.7)	Smaller files take up less space on HDDs, SSDs, etc.
Primary Memory (A1.1.4)	Smaller files move in and out of RAM faster

Connection	Why It Matters
Data Transfer	Smaller files transfer faster over networks

| This ties today's concept to prior ones rather than leaving it isolated.

Part 3: Deliberate Practice — ~20%

Activity A — Encode and Decode RLE Strings (Pairs):

Task	Instructions
Encode	BBBCCCCDDDEE → ?
Decode	5A2B4C1D → ?

Activity B — Classify File Types (Pairs):

For each file type, decide: Lossless or Lossy? Write a one-line justification.

File Type	Best Compression	Justification
Plain text (.txt)	Lossless	Any lost characters change meaning
Program code (.py, .java)	Lossless	One wrong byte stops execution
Photograph (.jpg)	Lossy	Small detail loss is unnoticeable
Song (.mp3)	Lossy	Imperceptible frequencies can be discarded
Simple icon (.png)	Lossless	Sharp edges and transparency must be preserved
Video (.mp4)	Lossy	Large file; imperceptible detail can be discarded

Part 4: Exit Check — ~10%

Two Items (Readable in Minutes):

Question	Purpose
Part A: "Outline one difference between lossy and lossless compression"	Checks understanding of defining properties
Part B: "Apply RLE to AAABBCCCCDDDDD and state the character count before and after"	Checks procedural application

Expected Answers:

Part A	Lossy permanently discards data; lossless rebuilds the original exactly
Part B	3A2B4C5D (14→8 characters)

Gap to Watch For: Defining lossy as "lower quality" rather than by **permanent discard**. Surface this before the next lesson.

3. Summary: Compression

Lossless vs. Lossy

Feature	Lossless	Lossy
Original can be rebuilt?	Yes — exactly	No — permanently discarded
File size reduction	Moderate	High
Quality	Preserved	Some loss (hopefully imperceptible)
Examples	RLE, ZIP, PNG, GIF	JPEG, MP3, MP4
Suitable for	Text, code, medical images	Photos, audio, video

Run-Length Encoding (RLE)

Feature	Description
Type	Lossless
Method	Replace runs of identical values with a count and the value
Works best on	Data with long repeated sequences
Fails on	Data with no repeats (increases file size)
Example	AAAABBBCCDAA → 4A3B2C1D2A

Transform Coding

Feature	Description
Type	Lossy
Method	Convert data to frequencies; discard imperceptible components
Examples	JPEG (images), MP3 (audio)
Key idea	Remove what humans cannot perceive

4. Differentiation & Extension

Support (Scaffolds That Can Be Removed)

Support Strategy	Description
Partially Completed RLE	Runs already bracketed; student supplies only the counts
Word Bank	For lossless/lossy comparison

Support Strategy	Description
Sentence Frames	From the EAL scaffold

Withdraw these as students gain fluency rather than leaving them permanently.

Challenge (Wider and Deeper, at the Same Command Band)

Challenge Strategy	Description
RLE Failure Case	Give a string on which RLE increases the size; describe why it behaves that way
Justification Practice	Describe one file type for which each method is the better choice with a justification

Keep at Describe — widening application without moving above describe into evaluating algorithms.

5. Engagement Activities (Supplementary)

If time allows, these can replace or supplement the main sequence:

Activity	Description
"Balloon Compression" Demonstration	Inflate a balloon (data) and slowly let air out (compression)—introduces lossy compression
"Texting Challenge"	Write a message using as few characters as possible (abbreviations, emojis)
"Image Compression Experiment"	Use online tools (TinyPNG, Compressor.io) to compress images and observe differences
"RLE Challenge"	Groups apply RLE to different strings and compare results

Activity	Description
"Compression Debate"	Team Lossy vs. Team Lossless debate advantages/disadvantages
"Ethical Compression Dilemma"	Discuss risks of compressing medical images, historical documents

6. Learning Objectives (Summary)

Knowledge and Understanding

- Define compression and explain its purpose in computer systems
- Differentiate between lossy and lossless compression methods
- Explain and apply run-length encoding (RLE) as a simple lossless compression technique
- Describe transform coding and its application in lossy compression methods

Skills

- Analyse data and identify opportunities for compression using different techniques
- Evaluate the suitability of different compression methods for various data types and applications
- Apply compression techniques to real-world examples and analyse the results

ATL Skills

Skill	Description
Communication	Effectively communicate understanding through discussions and written explanations
Collaboration	Work collaboratively to analyse data, apply techniques, and evaluate results
Critical Thinking	Analyse trade-offs between compression ratio, data quality, and processing requirements

Skill	Description
Information Literacy	Use online resources to deepen understanding
Reflection	Reflect on impact of compression on daily life

7. Resources

- Presentation (Slide Deck)
- InThinking eBook (A 1.1.8 with quiz)
- Worksheet: RLE practice + classification table
- Worksheet Answers

Author's Reflection

"The command term is Describe, so this lesson gives a detailed account and stops there. The earlier version reached into evaluating methods, analysing data for compression opportunities and even designing an algorithm, alongside a debate and an ethics dilemma. It worked well in class, but none of it earns marks at Describe, and it costs time this block cannot spare, so I have cut it back to the concept, the lossy and lossless distinction, and the two named techniques the syllabus actually asks for.

I have shaped the lesson around conceptual understanding rather than opening cold with a definition. This time the lesson starts with one file that visibly shrinks—the four-minute song that drops from about 40 MB to about 4 MB—drawing the idea of removing redundancy from that single case, then generalises before meeting RLE and transform coding as two instances of the same idea. The RLE failure case, where ABCD doubles in length, shows students where a method stops helping, not assuming compression always shrinks. The linking task at the end pulls compression back to storage and memory from earlier in the block, so it does not sit as an isolated topic.

I removed the mathematics from transform coding. Stronger students will want the transform itself, and I would rather point them to wider reading than spend the block's time on content that carries no return at Describe."